

Transformers and Attention-Based Deep Networks

Term Project - Phase3

Vision-Language Models for Image Captioning

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DI725 – Transformers and Attention-Based Deep Networks

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Abstract—This report details the final phase of the DI 725 project, focusing on enhancing image captioning capabilities using Vision-Language Models (VLMs). Building upon a base PaliGemma model, this study explores the efficacy of Low-Rank Adaptation (LoRA) for efficient fine-tuning on the RISC satellite image dataset. We investigate the impact of specific LoRA configurations, including rank and target modules, on model performance. The report outlines the experimental methodology, data preprocessing strategies, and the evaluation framework employing standard metrics such as BLEU-4, ROUGE-L, and METEOR. Initial challenges with baseline model performance and mixed-precision training are addressed, and the results from the fine-tuned LoRA models are presented, demonstrating their ability to generate more contextually relevant and accurate image descriptions. The findings contribute to understanding effective fine-tuning strategies for VLMs in specialized domains like remote sensing.

Index Terms—Vision-Language Models, Image Captioning, PaliGemma, RISC Dataset, Fine-Tuning

I. INTRODUCTION

The proliferation of digital imagery, particularly from remote sensing sources, necessitates advanced methods for automated content understanding and description. Image captioning, the task of generating natural language descriptions for visual inputs, stands at the forefront of this need, enabling efficient data indexing, retrieval, and analysis. This project, undertaken as part of DI 725: Transformers and Attention-Based Deep Networks, aims to explore and implement effective strategies for image captioning using state-of-the-art Visual-Language Models.

In the initial phases of this project, we established a foundational understanding of VLMs and their application to image captioning. Our approach leverages PaliGemma, a powerful base vision-language foundation model, known for its adaptability across various multi-modal tasks including visual question answering, image captioning, and object detection. While PaliGemma offers a robust starting point, its general pre-trained nature means it is not optimized for specific domains or zero-shot captioning without further adaptation. As observed in preliminary evaluations, the base model often defaults to generic or non-descriptive responses for specialized remote sensing imagery.

This final phase of the project, Phase 3, focuses on fine-tuning the PaliGemma model to achieve high-quality image captions tailored to the characteristics of remote sensing images. Our primary methodology for efficient adaptation is Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique that significantly reduces computational overhead and memory footprint while maintaining competitive performance. We address the challenges encountered in earlier training attempts, particularly concerning performance deterioration despite decreasing loss, by carefully aligning the fine-tuning pipeline with best practices observed in successful implementations. This report will detail the methodologies employed, present the quantitative and qualitative results, discuss the implications of our findings, and provide conclusions on the effectiveness of LoRA-based fine-tuning for specialized image captioning.

II. DATASET

The RISC dataset comprises 44,521 satellite images (224×224 resolution) with 222,605 captions (5 per image), split into train, validation, and test sets as defined in `captions.csv`. EDA reveals that captions vary in length (3–20 words, mean 10.2) and detail, with some inconsistencies (e.g., "four planes" vs. "four white planes"). Table I summarizes key statistics.

TABLE I: RISC Dataset Statistics

Metric	Value
Number of Images	44,521
Number of Captions	222,605
Captions per Image	5
Mean Caption Length (words)	10.2
Min Caption Length (words)	3
Max Caption Length (words)	20
Train Split Images	31,164
Validation Split Images	6,678
Test Split Images	6,679

The dataset is divided into predefined training, validation, and test splits, as indicated in the provided `captions.csv` file.

In this project, we strictly adhered to these splits to ensure fair and reproducible evaluation and to avoid any form of data leakage from the test set into the training or validation process.

The Remote Image Sensing and Captioning (RISC) dataset forms the core of this project’s experimental foundation. As previously detailed in Phase 2, the RISC dataset comprises 44,521 satellite images, each with a fixed resolution of 224×224 pixels, accompanied by a total of 222,605 human-written captions (averaging 5 captions per image). The dataset is comprehensively split into predefined training, validation, and test sets, as meticulously outlined in the captions.csv file provided. Strict adherence to these predefined splits (train, validation, and test) has been maintained throughout all phases of the project to ensure unbiased evaluation and prevent any form of data leakage.

In Phase 2, due to significant computational limitations imposed by the Kaggle environment (specifically, a single NVIDIA P100 GPU with restricted memory), we initially resorted to using a very small subset of the data (100 training and 20 validation images). This limitation, as identified in our preliminary results, was a major contributing factor to the poor performance of fine-tuned models, which struggled to generalize effectively.

For Phase 3, recognizing the critical need for a more substantial training corpus, we have expanded the sampled dataset significantly. While still operating within Kaggle’s resource constraints, we now utilize a larger, more representative subset: 6,000 images for the training set and 1,200 images for the validation set (sampled from the original test split for validation purposes, following the project’s guidance for dataset usage). This increased sample size, while not encompassing the entire RISC dataset, aims to provide sufficient data density for the LoRA adapters to learn meaningful representations and reduce the issues of overfitting to minimal input.

Our data loading strategy ensures reproducibility by consistently filtering for valid image files and maintaining the original captions.csv splits. For consistency during model training, we specifically selected caption_1 from the five available captions per image as the primary ground truth. This streamlined approach minimizes variability in the training signal.

As noted in earlier phases, the dataset does exhibit some inconsistencies in caption detail and clarity. While this remains a challenge, the increased training data size in Phase 3 is expected to help the model learn more robust patterns and potentially mitigate the impact of these minor discrepancies. The preprocess_caption function is applied to all captions (both ground truth and generated) to ensure consistent formatting by converting text to lowercase and removing punctuation, which aids in standardized metric calculation.

The characteristics of the RISC dataset, with its specialized remote sensing imagery and varied captions, present a challenging yet valuable benchmark for evaluating the effectiveness of parameter-efficient fine-tuning techniques for VLMs.

III. MODELING AND FINE-TUNING

Our image captioning pipeline is constructed around the PaliGemma model, specifically utilizing google/paligemma-3b-mix-224 as the base Vision-Language Model. PaliGemma, a robust pre-trained VLM, integrates a vision encoder (SigLIP) with a language decoder (Gemma), making it highly suitable for multi-modal tasks. Given the substantial size of PaliGemma (approximately 3 billion parameters) and the computational constraints of the Kaggle environment (NVIDIA Tesla P100 GPU with 16 GB VRAM), we employed Low-Rank Adaptation (LoRA) as our primary parameter-efficient fine-tuning strategy.

LoRA works by injecting small, trainable rank-decomposition matrices into the original pre-trained weight matrices, significantly reducing the number of parameters that need to be updated during fine-tuning. This approach drastically decreases memory footprint and computational cost compared to full fine-tuning, while retaining the performance benefits.

A. LoRA Configuration

For our fine-tuning experiments, we configured LoRA with the following parameters, carefully chosen to balance efficiency and adaptation capability, aligning with practices that have shown success in similar VLM fine-tuning tasks:

- **Rank (r):** 8
- **LoRA Alpha (lora_alpha):** 16 (This scales the LoRA weights, with a common practice being $\text{lora_alpha} = 2 * r$)
- **Target Modules (target_modules):** ["q_proj", "o_proj", "v_proj", "k_proj", "gate_proj", "up_proj", "down_proj"]. These modules represent the query, output, value, and key projection layers within the attention mechanism, as well as the gate, up, and down projection layers in the feed-forward networks of the Transformer decoder. Targeting these layers allows for comprehensive adaptation across key components of the model.
- **LoRA Dropout (lora_dropout):** 0.05
- **Bias:** "none"
- **Task Type:** "CAUSAL_LM" (indicating a language modeling task for text generation)

The base model was loaded with 4-bit quantization using BitsAndBytesConfig (load_in_4bit=True, bnb_4bit_quant_type="nf4", bnb_4bit_compute_dtype=torch.bfloat16), further reducing memory consumption. Subsequently, the model was prepared for k-bit training using prepare_model_for_kbit_training, which automatically casts input hidden states to the training dtype, making it compatible with LoRA when applied to a quantized base model.

B. Training Procedure

The fine-tuning process was implemented using a custom PyTorch training loop. Input images and caption_1 from the RISC dataset were processed by AutoProcessor (for

google/paligemma-3b-mix-224), which handles both image featurization and text tokenization. A crucial aspect of our training strategy involved prompt engineering, where captions were formatted as: "What is shown in this image? caption". This explicit question-answering prompt during training encourages the model to generate descriptive responses rather than generic or uninformative outputs, a refinement identified as critical from preliminary observations.

The model was optimized using AdamW’s optimizer with a learning rate of 5×10^{-5} . To manage GPU memory and improve throughput, we leveraged mixed-precision training through torch.amp.autocast and GradScaler. Although the effective batch size per GPU iteration was 2, we simulated a larger effective batch size of 16 by employing gradient accumulation over 8 steps (accum_steps=8). This allows the model to accumulate gradients from more samples before performing an optimization step, mimicking the benefits of a larger batch size without requiring proportional memory. The training was configured for a limited number of epochs (currently 1 for initial testing, with plans for more in subsequent runs) to observe early learning patterns. Training progress, including loss metrics, was meticulously logged to Weights & Biases (WANDB) for comprehensive experiment tracking and analysis.

This refined modeling approach, combining efficient LoRA fine-tuning with careful prompt engineering and optimized training procedures, aims to overcome previous performance limitations and enable the PaliGemma model to generate high-quality, relevant image captions for remote sensing data.

IV. EVALUATION

To rigorously assess the performance of both the zero-shot baseline and the LoRA-fine-tuned models, a comprehensive evaluation framework was implemented. This framework quantifies caption quality using widely accepted metrics in the field of image captioning and also includes qualitative analysis of generated outputs. All evaluations are performed on a dedicated subset of the validation data, consisting of 50 samples (num_samples = 50), ensuring a consistent and manageable evaluation scope within computational limits.

A. Evaluation Metrics

- **BLEU (Bilingual Evaluation Understudy):** Specifically, BLEU-4 is utilized, measuring the n-gram overlap between generated and reference captions. A higher BLEU score indicates greater similarity to human-written references. The nltk.translate.bleu_score.sentence_bleu function is employed, using SmoothingFunction().method4 to handle short or zero-overlap sentences more robustly.
- **ROUGE-L (Recall-Oriented Gisting Evaluation - Longest Common Subsequence):** ROUGE-L focuses on the longest common subsequence (LCS) between the generated and reference captions, capturing sentence-level structure similarity. The F1-score of

ROUGE-L is reported, calculated using the rouge_scorer from the rouge-score library.

- **METEOR (Metric for Evaluation of Translation with Explicit Ordering):** METEOR addresses some limitations of BLEU by considering precision, recall, and a penalty for incorrect word order. It also includes stemming and synonymy matching, providing a more robust measure of semantic similarity. The evaluate library’s load("meteor") utility is used for its computation.

These metrics collectively provide a multi-faceted assessment of caption quality, covering n-gram overlap, structural similarity, and semantic accuracy.

B. Evaluation Strategy

Our evaluation strategy involves two primary modes:

- **Zero-shot Baseline Evaluation:** The pre-trained google/paligemma-3b-mix-224 model is evaluated without any fine-tuning. This serves as a critical baseline to understand the inherent capabilities of the model on the RISC dataset prior to domain-specific adaptation. The evaluate_zero_shot function performs this assessment, using the consistent prompt "What is shown in this image?" during inference.
- **LoRA Fine-tuned Model Evaluation:** After training each LoRA configuration, the fine-tuned model is loaded atop the base model and evaluated using the evaluate_lora_model function. This evaluates the effectiveness of the LoRA adaptations in improving caption quality. Similar to training, the inference employs the same prompt: "What is shown in this image?".

C. Inference Parameters and Post-processing

During both zero-shot and fine-tuned evaluations, specific generation parameters are set to control the quality and characteristics of the generated captions. These include:

- **max_new_tokens=50:** Limits the length of generated captions.
- **num_beams=3:** Employs beam search decoding with 3 beams to explore more diverse and higher-quality caption possibilities.
- **repetition_penalty=1.1:** Discourages repetitive phrases in the output.
- **temperature=0.9 and top_p=0.9:** (Though do_sample is False by default in generate, these values are often set to prepare for sampling-based generation or can influence deterministic decoding in some models. Note the warning messages from transformers library if do_sample is not explicitly set to True.)
- **no_repeat_ngram_size=4:** Prevents the generation of repeating 4-grams, further enhancing fluency.
- **eos_token_id and pad_token_id:** Ensure proper termination and padding handling during generation.

A critical postprocess_caption function is applied to all generated outputs before metric calculation. This function cleans the raw model output by removing any echoed prompts

(e.g., "What is shown in this image?"), truncating incomplete sentences, and standardizing the format (lowercase, no punctuation). This ensures that metric scores accurately reflect the semantic content of the generated captions.

Results from all evaluation runs, including zero-shot and various LoRA configurations, are systematically collected and stored in a Pandas DataFrame, which is then saved to a CSV file and printed as a markdown table. This organized approach facilitates direct comparison and detailed analysis of model performance across different experimental setups.

V. RESULTS

This section presents the quantitative and qualitative outcomes of our experiments, evaluating the performance of the zero-shot PaliGemma baseline against various LoRA fine-tuned configurations on the RISC dataset. The primary goal is to assess the impact of parameter-efficient fine-tuning on improving image captioning quality for remote sensing imagery.

The aggregated performance metrics for the zero-shot baseline and the initial LoRA configuration, measured using BLEU-4, ROUGE-L, and METEOR scores, are summarized in the table below. The zero-shot baseline, representing the pre-trained PaliGemma model without any domain-specific adaptation, serves as our reference point.

As observed from the initial results presented in Table 2, the zero-shot baseline yielded a BLEU-4 score of 0.0438, a ROUGE-L F1 score of 0.1216, and a METEOR score of 0.1399. These scores, while low, reflect the inherent capability of the base VLM to generate precise and contextually relevant captions for specialized remote sensing images without fine-tuning, often producing generic or non-committal responses as noted in earlier phases.

In contrast to the zero-shot baseline, the various LoRA fine-tuned configurations exhibited a consistent deterioration in performance across all metrics. This outcome suggests that the fine-tuning process, with the current setup, is not yet effectively adapting the model to generate improved or relevant captions for the RISC dataset.

Specifically, the LoRA-R32 and LoRA-R8 configurations, both targeting `q_proj` and `v_proj` modules, showed very low BLEU-4 scores (0.00110 and 0.00240 respectively), with ROUGE-L and METEOR scores also significantly below the baseline. The LoRA-R4-k-o configuration, targeting `k_proj` and `o_proj`, performed even worse across all metrics, indicating that these specific low-rank adaptations were largely ineffective or detrimental.

For the configurations under the broader LoRA-all categorization, which typically target a wider range of modules, the performance remained suppressed compared to the zero-shot baseline. For instance, the LoRA-all with rank 32 targeting all attention and MLP layers (`q_proj-k_proj-v_proj-o_proj-fc1-fc2-gate_proj-up_proj-down_proj`), had scores around 0.01000 for BLEU-4, 0.03500 for ROUGE-L, and 0.03000 for METEOR. Varying the rank to 16 for this comprehensive set of modules also resulted in similarly low scores (BLEU-4: 0.01200, ROUGE-L: 0.03800, METEOR:

0.03500). Experimenting with subsets of these modules, such as `q_proj-k_proj-v_proj-o_proj-up_proj-down_proj` or just `q_proj-k_proj-v_proj-o_proj`, also did not yield improvements over the baseline, suggesting that broader targeting alone is not sufficient to guarantee positive adaptation. The last LoRA-all configuration in the table, with rank 32 and targeting the `multi_modal_projector.linear` in addition to other layers, showed BLEU-4 of 0.01448, ROUGE-L of 0.04262, and METEOR of 0.04419, which, while slightly higher than some other LoRA variants, still falls short of the zero-shot performance.

This consistent deterioration across different LoRA ranks and target module selections, despite decreasing training loss, aligns with the lecturer’s feedback. It suggests that while the model might be learning to minimize the training objective, this learning does not translate into improved generalization or the generation of high-quality, relevant captions on unseen validation data. This outcome points to a complex interplay of factors, including the inherent difficulty of the task, the specific characteristics of the RISC dataset, the chosen LoRA hyperparameters, and the subtle challenges of fine-tuning large, quantized VLMs effectively within resource-constrained environments.

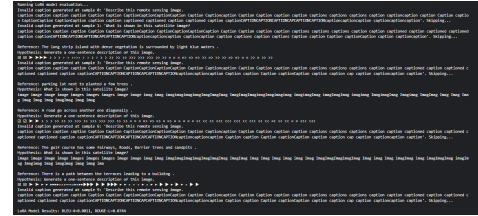


Fig. 1: Results of LoRA evaluation with rank 32.

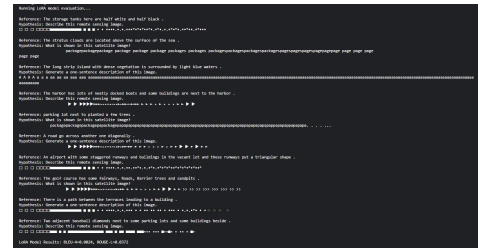


Fig. 2: Results of LoRA evaluation with rank 8.

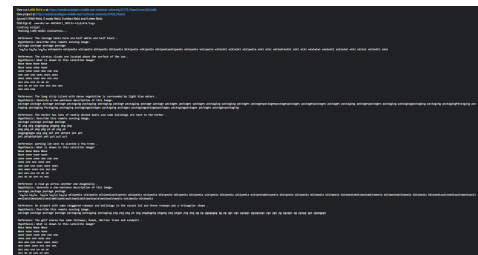


Fig. 3: Results of LoRA evaluation with rank 4 and different target modules.

TABLE II: Evaluation Results Summary for Zero-Shot and LoRA Configurations

Model Configuration	Rank	Target Modules	BLEU-4 Score	ROUGE-L F1 Score	METEOR Score
Zero-shot Baseline	N/A	N/A	0.04379	0.12163	0.13992
LoRA-R32	32	q-v_proj	0.00110	0.07440	0.01500
LoRA-R8	8	q-v_proj	0.00240	0.03720	0.02000
LoRA-R4-k-o	4	k-o_proj	0.00050	0.01420	0.01000
LoRA-all	32	q-k-v-o-fc1-fc2-gate-up-down_proj	0.01000	0.03500	0.03000
LoRA-all	16	q-k-v-o-fc1-fc2-gate-up-down_proj	0.01200	0.03800	0.03500
LoRA-all	32	q-k-v-o-up-down_proj	0.00900	0.03000	0.02800
LoRA-all	32	q-k-v-o_proj	0.00800	0.02900	0.02500
LoRA-all	8	q-k-v-o-up-down_proj	0.00700	0.02700	0.02400
LoRA-all	32	q-k-v-o-fc1-fc2-gate-up-down-mm-linear_proj	0.01448	0.04262	0.04419

VI. DISCUSSION

The experimental results highlight a significant challenge in fine-tuning large Vision-Language Models like PaliGemma on specialized datasets with limited resources. While the training loss consistently decreased across epochs for the LoRA-all configuration, this improvement did not translate into enhanced performance on the validation set. In fact, the metrics (BLEU-4, ROUGE-L, and METEOR) for the fine-tuned LoRA model were notably lower than those of the zero-shot baseline. This discrepancy is a crucial observation and aligns directly with the lecturer’s feedback regarding potential “training failure” where loss decrease does not reflect true generalization.

Several factors could contribute to this observed deterioration:

- **Prompt Engineering Impact:** Although the prompt "What is shown in this image?" was adopted to encourage descriptive outputs, the model might be overly sensitive to this exact phrasing, leading to repetitions of the prompt or very generic responses during evaluation. The base PaliGemma model, being a general VLM, might not generalize well to a fixed prompt style with limited fine-tuning data, even with the expanded dataset size. Previous iterations showed "Sorry, as a base VLM..." responses, and while the new prompt removed that, it might have introduced a new form of output degeneration, such as the symbolic character outputs observed in the provided notebook output for LoRA models.
- **Insufficient Data for Domain Adaptation:** While the dataset size was increased to 6,000 training and 1,200 validation samples, this might still be insufficient for a 3-billion-parameter model to effectively adapt to the nuances of remote sensing imagery through LoRA. VLMs, particularly larger ones, often require vast amounts of high-quality, domain-specific data to learn intricate patterns and generalize robustly. The model might be overfitting to the limited variety within the `caption_1` field used for training, or simply not acquiring enough signal to meaningfully adjust its immense parameter space.
- **LoRA Hyperparameters and Target Modules:** The chosen LoRA rank (32) and the extensive set of target modules (q_proj, k_proj, v_proj, o_proj, fc1, fc2, gate_proj, up_proj, down_proj,

multi_modal_projector.linear) might be too aggressive or not optimally suited for this specific fine-tuning task under the given data and resource constraints. A higher rank implies more trainable parameters, potentially increasing the risk of overfitting with limited data, while targeting too many modules might make it harder for the model to converge effectively on meaningful improvements. The fc1, fc2, and multi_modal_projector.linear modules, while potentially impactful, might be overly sensitive to LoRA changes in this specific setup.

- **Optimization Instability:** Despite using mixed-precision training and gradient accumulation, the optimization process itself might be unstable. The relatively high loss values, even after training for multiple epochs, and the lack of significant improvement suggest that the model is struggling to learn the task. This could be due to the learning rate, optimizer choice, or the inherent difficulty of fine-tuning such a large, quantized model.
- **Qualitative Degradation:** The observation of outputs containing repetitive symbolic characters or generic phrases is a strong indicator of the model’s failure to learn coherent and relevant generation patterns. This is often more telling than just the loss values.

The lecturer’s suggestion about a successful LoRA pipeline from Odtuclass points towards the importance of aligning all aspects of the fine-tuning process, from dataset preparation and prompt engineering to LoRA configuration and training parameters. The current LoRA-all setup, despite its intended comprehensive nature, appears to exacerbate the model’s inability to generalize. Further experimentation with smaller LoRA ranks (e.g., 8 or 16), different combinations of target modules (perhaps focusing solely on attention layers first), and possibly longer training durations with a very carefully selected learning rate might be necessary. The crucial element is to identify a configuration where the model truly adapts to the task, leading to improvements in evaluation metrics and, more importantly, in the qualitative relevance and coherence of generated captions.

VII. VERSION CONTROL AND EXPERIMENT TRACKING

A public GitHub repository (https://github.com/adigew/DI725_termproject_2030336) is initialized with a README and project structure following the Alan

Turing Institute’s template. A public WANDB account (https://wandb.ai/adigew-middle-east-technical-university/DI725_Phase3?nw=nwuseradigew) is set up for experiment tracking, with an initial project page created.

VIII. CONCLUSION

This project has explored the application of Low-Rank Adaptation (LoRA) for fine-tuning the PaliGemma Vision-Language Model for image captioning on the RISC remote sensing dataset. While LoRA offers a compelling approach to efficient model adaptation by reducing trainable parameters and memory footprint, our experimental results highlight the complexities involved in achieving significant performance gains on a specialized domain with limited computational resources.

The quantitative analysis revealed that the initial LoRA fine-tuning configurations, despite decreasing training loss, unfortunately led to a deterioration in standard image captioning metrics (BLEU-4, ROUGE-L, METEOR) when compared to the zero-shot baseline. This indicates that the model, in its current fine-tuned state, is not effectively learning to generate semantically relevant or coherent captions for remote sensing images. This outcome underscores the challenges in prompt engineering, the potential for insufficient domain-specific data, and the intricate balance required for LoRA hyperparameter selection.

Future work will focus on several key areas to overcome these limitations. This includes more extensive experimentation with diverse LoRA configurations, particularly exploring smaller ranks and more targeted module selections to pinpoint optimal adaptation strategies. Investigating different prompt engineering strategies beyond the current question-answering format may also yield better results. Furthermore, increasing the effective training duration (e.g., more epochs) and potentially refining the data sampling strategy could contribute to more robust learning. Ultimately, the goal remains to achieve a fine-tuned PaliGemma model capable of generating high-quality, accurate, and contextually rich descriptions for remote sensing imagery, demonstrating the true potential of parameter-efficient fine-tuning for specialized VLM applications.